

Austin Fairbanks

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Education

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| University of Massachusetts Amherst , BS in Computer Science | Sept 2023 – May 2026 |
| University of Massachusetts Amherst , MS in Computer Science (Focus in Data Science) | Sept 2026 – May 2027 |
- GPA: 4.0/4.0
 - **Coursework:** Machine Learning (CS 589), Algorithms for Data Science (CS 514), Introduction to Algorithms (CS 311), Introduction to Machine Learning (CS 389)

Publications

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| Benchmarking of DataFrame Libraries in Applications of Data-Driven Modeling | Summer 2025 |
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- Conducted benchmarking of dataframe libraries (Polars, cuDF, Pandas) for applied modeling in manufacturing
 - Investigated internal memory layouts, indexing strategies, and columnar performance tradeoffs across libraries
 - Built an interactive Streamlit app to present results and recommend optimal frameworks based on task types
 - Tech Stack: Polars, cuDF, Pandas, Streamlit
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| Hyperparameter Optimization Manual | Summer 2024 |
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- Authored a comparative manual covering setup, tuning behavior, and real-world use of Optuna, Keras Tuner, and Scikit-learn's tuning API
 - Explained advanced local functionality that lacked official documentation
 - Tech Stack: Python, TensorFlow, Scikit-learn, Optuna, Keras Tuner

Experience

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| Machine Learning Engineering Intern (via Magnit) , Corning Inc. – Corning, NY | May 2025 – Present |
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- Benchmarked GPU-accelerated dataframe libraries to reduce I/O bottlenecks in high-volume modeling pipelines
 - Developed custom resampling strategies for multimodal sensor data used for data-driven process modeling
 - Trained CV models using transfer learning to detect machine faults from thermal and visible imaging
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| Machine Learning Intern (via Magnit) , Corning Inc. – Corning, NY | July 2024 – Aug 2024 |
|--|----------------------|
- Wrote a series of internal guides on practical HPO using Optuna and Keras Tuner
 - Built preprocessing functions and scalable pipelines for internal CV project
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| Intramural Program Assistant , UMass RecWell – Amherst, MA | Aug 2025 – Present |
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- Progressed through leadership roles from Referee to Manager to Program Assistant
 - Managed logistics and scheduling for 40+ student referees
 - Trained and mentored new officials while mediating disputes and serving as liaison with upper-level staff

Projects

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| RateMyProfessors Stats | Jan 2025 – Present |
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- Scraped and compiled over 5 million student reviews using GraphQL, published on Kaggle
 - Created a full-stack dashboard to visualize trends using Next.js, MongoDB, and Typescript
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| OnePrompted: Prompt Engineering with Vertex AI | Apr 2025 |
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- Built a lightweight middle-layer model trained to rewrite and refine vague prompts
 - Demonstrated meaningful improvements in response quality using pairwise evaluations
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| Technical Blog | Ongoing |
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- I regularly publish articles exploring data science, machine learning, and optimization in applied settings.

Skills

Languages: Python, C++ , SQL, Java, HTML, JavaScript/TypeScript
ML/DS Libraries: PyTorch, TensorFlow, Scikit-learn, Optuna, Pandas, cuDF, Polars
Tools: Jupyter, Keras Tuner, Streamlit, LaTeX